# Candi Zheng

+86 18038048021 • czhengac@connect.ust.hk

## Education

- **Ph.D. Candidate in Mathematics**, Feb 2019 – Present, expected graduation date early 2024
  The Hong Kong University of Science and Technology
- **Ph.D. Student in Mechanical and Aerospace Engineering**, Feb 2017 – Present
  The Southern University of Science and Technology (Joint Program)
- **Ph.D. Student in Mechanical and Aerospace Engineering**, Sep 2017 – Feb 2019
  The Hong Kong University of Science and Technology (Joint Program)
- **Visiting Student**, Sep 2016 – Jan 2017
  The University of Edinburgh
- **B.S. in Physics**, Sep 2013 – Jun 2017
  The Southern University of Science and Technology

## Research Interest

My aim is to significantly improve understanding and predictive power in both the fields of AI and physics. I specialize in designing AI models for non-equilibrium physics systems, focusing on models that provide distinct physical insights and enhance predictive performance. Furthermore, I'm particularly interested in how insights from non-equilibrium physics can enhance generative AI models, particularly those grounded in physics such as diffusion models.

## Publications

1. **Candi Zheng***, Yuan Lan. "Characteristic Guidance: Non-linear Correction for DDPM at Large Guidance Scale." cs.CV, arXiv:2312.07586 (2023).
2. **Candi Zheng***, Yang Wang, and Shiyi Chen. "Phase Transition in Extended Thermodynamics Triggers Sub-shocks." physics.flu-dyn, arXiv:2304.10742 (2023).
3. **Candi Zheng***, Yang Wang , and Shiyi Chen. "Stabilizing the Maximal Entropy Moment Method for Rarefied Gas Dynamics at Single-Precision." physics.flu-dyn, arXiv:2303.02898 (2023).
4. **Candi Zheng***, Yang Wang, and Shiyi Chen. "Data-driven constitutive relation reveals scaling law for hydrodynamic transport coefficients." Physical Review E 107, no. 1 (2023): 015104.
5. Jin, Yuan-Jun, Rui Wang, Jin-Zhu Zhao, Yong-Ping Du, **Can-Di Zheng**, Li-Yong Gan, Jun-Feng Liu, Hu Xu*, and S. Y. Tong*. "The prediction of a family group of two-dimensional node-line semimetals." Nanoscale 9, no. 35 (2017): 13112-13118.

* for corresponding author

## Patents

- Yang Wang, **Candi Zheng**, Hongli Li, Shiyi Chen. "Machine Learning Early Risk Assessment Model and Charts for Unstable Angina Patients' ICA timing strategy." US Provisional Patent Application No. 63/401,721

## Awards and Scholarships

- **Risk Classification Top 10**, ATEC 2018 Developer AI Challenge, Primary round, 2018
- **Outstanding Graduation Project**, Southern University of Science and Technology, 2017
- **The University Physics Competition**, Silver, 2015
- **RoboMaster**, Central South China Division, Third Class, 2015
- **National Encouragement Scholarship**, Southern University of Science and Technology, 2014
- **Outstanding Freshman Scholarship**, Southern University of Science and Technology, 2014

## Experience

- **Clinical Research Coordinator (Part time)**, July 2018 – July 2021
  Shanghai General Hospital, medical AI algorithm designing.

## Skills

- **Language:** English, Chinese
- **Math:** Qualified as Ph.D candidate in pure mathematics options (analysis and algebra).
- **Physics:** Focusing on in statistical physics and kinetic theory of gas. Some research experience on first-principle simulation in condensed-matter theory.
- **Programming:** Proficient in Python for a diverse range of applications. My github project for maximal entropy moment method: MomentGauge. Some experience with JAVA, Fortran, CUDA C, and MPI for high-performance computing tasks.
- **Machine Learning:** Proficient in common machine learning frameworks and tools, including Pytorch, Jax, Scikit-learn.